

Operating Systems : Introduction Explore the basic of operating systems and core features that power your computer.

Task 1:- Introduction

You turn on your phone or laptop every day, and everything just works: Apps launch, files open, music plays, and whole system feels seamless. But what actually makes all possible?

Now its time to uncover the **Invisible layer** that ties it all together: **The operating system OS.**

Scenario:-

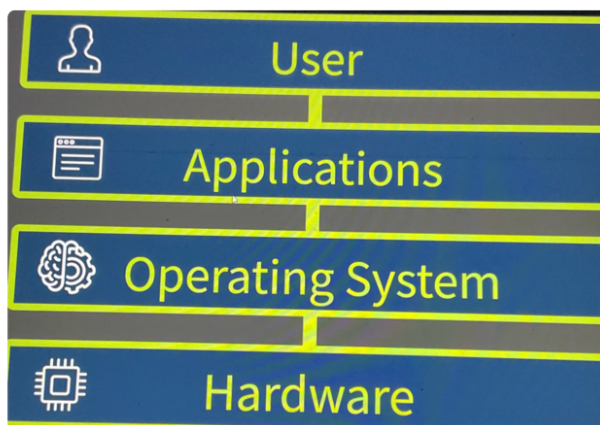
Your friend just upgraded their primary computer and knowing, your getting into tech, has generously gifted you their old machine. Its been untouched for a while, and he can't remember much about it, only that it used to run well. Before you can decide whether to upgrade it, wipe it, sell it or turn it into your next project, you need to figure out precisely what you're dealing with.

Learning Objectives:-

- * Understand what an operating system is and the role it plays
- * Explain the core duties of an operating system.
- * Identify common OS types and their typical use cases
- * Practice interacting with OS to gather system information.

Task 2:- The Invisible Manager

* An operating system (OS) is the core software that coordinates everything happening on a computer. It sits between the user, applications and system's physical hardware, acting as **Invisible Manager** that keeps entire machine running as one unified system.



A helpful Analogy is think computer as busy Airport, which all its components functioning together.

- * Your Hardware (CPU, RAM, Storage, connected device): The runways, airplanes, fuel systems, radar, and other physical infrastructure.
- * Your Application (web browser, game launcher): The various passengers, all trying to take off, land etc.
- * Your Operating System (Windows, Linux, MacOS): The entire air traffic control system, directing all of this activity.

We need an operating system because it provides this all important job coordination and structuring that makes modern computer possible. Without OS each computer need direct control over CPU, memory, files, devices and security. This would quickly conflict, and OS handles this by acting as central organizer.

System privilege layers: -

Inside a modern computer different part of system operate at various permission levels. Some components directly communicate with hardware, while regular applications run is safer. This separation is intentional and help prevent conflicts and security issues.

* Kernel Space: - The privileged, locked down core of OS. This is where kernel, part of OS that directly manages hardware and system resources, runs. It has unrestricted access to CPU, memory, storage and all hardware components.

* User Space: - where all standard application runs. Applications in user space are deliberately prevented from accessing hardware directly. Whenever they need to open or save a file, play a sound or connect to wifi, they must make a system call and request, that kernel act on their behalf.

In airport analogy: -

Kernel space is the control tower, they alone can directly control runways, radar and other hardware.

Application is user space like passengers, Airlines.

Operating System duties: -

Now that you know what an operating system is how system privilege is separated. Every OS is responsible for few core duties that allow your computer to run safely, efficiently and predictably.

OS Responsibility	What the OS Does	Example
Process Management	Creates, schedules, prioritizes, and terminates running programs. The OS decides how much CPU time each process gets, making multitasking feel seamless	Opening multiple apps, like your browser, music player, and social media, without your computer freezing
Memory Management	Allocates RAM to processes, protects the app's memory from other processes, and reclaims memory when apps are closed. When RAM runs low, the OS uses virtual memory to keep your system stable	Opening multiple app at once, the OS allocates RAM to each one and keeps them isolated so they don't interfere or crash each other
File System Management	Organizes files into directories, handles naming, paths, permissions, metadata (name, size, type, timestamps)	Creating a new folder, saving a photo, or setting a file to "read only"
User Management	Handles multiple user accounts, authentication, and permissions to determine who can access what	Logging in with your password and keeping your files inaccessible to other user accounts

	access while	Unauthorized access to other user resources
Device Management	Loads drivers and provides a universal interface (hardware abstraction layer), so apps can say "print this" or "play this sound"	Plugging in a new mouse, printer, or external hard drive and having it work immediately

Operating System Security:—

It's important to understand that every OS also acts as Security foundation. Before any antivirus, firewall or security tool is introduced, The OS is already enforcing protection in background.

At basic level your Operating System handles:—

* **Authentication**:— verifies who you are through login passwords and biometrics.

* **Permissions**:— Controls exactly what each user/app is allowed to read, write or execute.

* **Isolation**:— Keeps every process in its own protected box (Kernel / user space separation).

* **System protection**:— Safeguards critical system files and settings from unauthorized changes.

Quiz:—

Which OS ^{space} has unrestricted access to your computer's hardware?

A) Kernel space

a) which OS responsibility manages user accounts, authentication and permissions?
A) User management

About this computer opened.



Q) After opening **About this computer** shortcut, you are greeted with an Overview of System's specification. What version of ubuntu mate is your computer running?

1.26.2

Q) check out Hardware section of System tab. How much memory is allocated to your machine.

A) 1.9 GiB